CAPSTONE 43

AI Study of Generative AI Impact on Jobs

A research-based college project with a Python demo

PROJECT QUESTION
What does credible research say about generative AI and jobs, and what should students do?

PREPARED FOR
College Project Submission

PROJECT TYPE
Research + Python Working Demo + Responsible-AI Verification

DATE PREPARED
October 2026

Important Submission Note: This report includes a ready-to-run prototype and interview questions. Conduct two real interviews and replace the interview-note placeholders with the interviewees' actual responses before submitting. Do not present invented responses as real research.

ABSTRACT

Generative artificial intelligence (GenAI) can create text, code, images, audio, and other content from prompts. It may help workers complete some tasks faster, change the skills employers expect, and create new work; it can also produce inaccurate information, expose private data, reproduce biased outputs, and put pressure on some routine tasks. This project examines the evidence from the World Economic Forum (WEF), the International Labour Organization (ILO), and peer-reviewed academic research. It compares optimistic and cautious interpretations, proposes an upskilling plan for students, and presents a small Python application that helps users explore how different job tasks may be affected. The central finding is that task-level change is more useful to study than assuming that entire occupations will simply disappear.

Keywords: generative AI, employment, job transformation, automation, augmentation, upskilling, Python, responsible AI.

1. INTRODUCTION

GenAI tools are increasingly used for drafting, summarising, coding assistance, brainstorming, translation, customer support, and data analysis. These uses can increase productivity, but the result depends on the task, the quality of the prompt, the available data, and human review. A model may produce confident but false claims, and a fast first draft is not necessarily a correct final answer.

This project treats AI as a technology that changes tasks within jobs. For example, a software developer may use AI to draft tests or explain code while remaining responsible for architecture, security, debugging, and decisions. A teacher may use AI to draft learning material while retaining responsibility for accuracy, student needs, and assessment.

2. AIM AND OBJECTIVES

- **Aim:** Investigate credible evidence about GenAI's possible effects on jobs and translate findings into practical steps for students.
- Collect and summarise reports from WEF and ILO, plus academic research; include an India-oriented industry perspective where verifiable.

3. RESEARCH METHODOLOGY

This is a secondary-research project. Sources were selected for institutional credibility, methodological transparency, and relevance to employment. The report uses the WEF Future of Jobs Report 2025, the ILO's 2025 update on GenAI and jobs, and a peer-reviewed experiment published in *Science*. Figures are reported with their original scope and date. Forecasts from employer surveys are treated as expectations, not guaranteed outcomes.

4. WHAT CREDIBLE RESEARCH SAYS

4.1 World Economic Forum (WEF), Future of Jobs Report 2025

The WEF report draws on a survey of more than 1,000 employers representing over 14 million workers across 55 economies. It estimates that broader labour-market trends could create 170 million jobs and displace 92 million by 2030, a net increase of 78 million. These figures cover several drivers—not GenAI alone—including technological change, demographic shifts, economic uncertainty, and the green transition. The report also says employers expect about 39% of workers' core skills to change by 2030.

Interpretation: Job growth and displacement happen simultaneously. Figures reflect employer expectations, not a guarantee for every region or occupation.

4.2 International Labour Organization (ILO), 2025 Update

The ILO's 2025 update estimates that around one in four workers globally is in an occupation with some degree of exposure to GenAI. It places 3.3% of global employment in the highest exposure category. The report emphasises that exposure does not automatically mean job loss: most occupations include tasks requiring human input, so transformation is generally considered more likely than complete replacement.

Interpretation: Exposure measures technical potential, not actual job removal. Adoption costs, regulation, customer expectations, data quality, and human accountability affect real-world outcomes.

4.3 Academic Evidence: Noy and Zhang (Science, 2023)

In a preregistered experiment involving 453 college-educated professionals completing mid-level writing tasks, participants who could use ChatGPT completed tasks about 40% faster on average and produced outputs rated about 18% higher in quality. This provides evidence that GenAI can improve performance on specific writing tasks, but it does not mean all jobs become 40% faster.

4.4 India and Industry Context

For an India-focused submission, include a current NASSCOM report or industry briefing only after opening the original publication and checking its date, methodology, and exact claim. Verified NASSCOM sources can be added using the source log template in Section 11.

> **Research Takeaway:** GenAI automates parts of work, assists with tasks, and creates demand for new skills. Students should learn to use AI critically while developing strong domain knowledge and practical capabilities.

5. OPTIMISTIC VS CAUTIOUS VIEWS

Dimension	Optimistic Interpretation	Cautious Interpretation	Balanced Conclusion
Employment	New products and services create roles and expand productivity.	Some routine tasks and roles may shrink; transitions can be uneven.	Prepare for role changes and monitor actual local demand.
Productivity	AI speeds up drafting, coding support, research, and summarisation.	Checking, rework, integration, and errors can erase time savings.	Measure the full workflow, including verification time.
Skills	AI literacy helps students learn and build projects faster.	Over-reliance can weaken fundamentals and make copied work obvious.	Use AI as a coach; explain and test everything yourself.
Reliability & Quality	Assistive tools help produce stronger first drafts and alternatives.	Outputs may be false, outdated, biased, or completely fabricated.	Verify all claims against primary reliable sources.

6. TASK-LEVEL IMPACT EXAMPLES

Occupation	Tasks AI May Assist With	Human Responsibilities Remaining
Software Developer	Code explanation, drafting tests, documentation, boilerplate code.	Requirements, architecture, security, debugging, final review.
Teacher / Trainer	Lesson outlines, quiz drafts, examples, adapting reading levels.	Accuracy, inclusion, student relationships, assessment, care.
Marketing / Comms	Brainstorming, drafting copy, summarising customer feedback.	Brand judgement, fact checking, consent, audience understanding.
Data Analyst	Drafting SQL or Python, explaining patterns, summarising reports.	Data quality, statistical validity, privacy, business interpretation.

7. RISKS, ETHICS AND RESPONSIBLE USE

- **Hallucinations:** Models can invent facts, citations, dates, or APIs. Verify every important claim.
- **Privacy:** Do not paste personal, confidential, or credential data into public AI tools without safeguards.
- **Bias & Fairness:** Outputs may reproduce stereotypes; test and review consequential decisions with humans.
- **Copyright & Disclosure:** Respect intellectual property and state clearly where AI materially helped.

8. STUDENT UPSKILLING PLAN (8 WEEKS)

Week	Learning Focus	Practical Deliverable
Week 1	AI foundations: capabilities, limits & risks.	One-page concept note and glossary.
Week 2	Prompt design: context, constraints & formatting.	Prompt library with 5 prompts & revisions.
Week 3	Python fundamentals and data handling.	Script reading and summarising CSV data.
Week 4	AI-assisted coding, testing and debugging.	Program with test cases & documented fixes.
Week 5	Research literacy and source verification.	Evidence table with 5 verified primary claims.
Week 6	Visual communication & data storytelling.	One-page infographic of verified statistics.
Week 7	Responsible AI, privacy, copyright & ethics.	Risk checklist and AI-use disclosure statement.
Week 8	Portfolio development and demo practice.	Working demo, README, and presentation slides.

9. INTERVIEW PLAN: TWO LOCAL PROFESSIONALS

The assignment requires two real interviews (one teacher/faculty member and one working professional). Record actual responses before final submission.

Field	Interview 1 (Faculty)	Interview 2 (Professional)
Role / Sector	*[Fill in after interview]*	*[Fill in after interview]*
AI Tasks Used	*[Actual response notes]*	*[Actual response notes]*
Benefits & Risks	*[Actual response notes]*	*[Actual response notes]*
Advice to Students	*[Actual response notes]*	*[Actual response notes]*

10. PYTHON WORKING PROJECT: GENAI JOB TASK EXPLORER

The prototype is a small, rule-based Python application. A user selects a sample task and sees how AI may assist, what a human should verify, and which skills to build. It uses standard library tools and requires no paid API keys.

```
      TASKS = { "1": { "role": "Software Developer", "task": "Drafting unit tests or code doc
umentation", "category": "Often assistive", "ai_help": "Suggest test cases, explain code, dra
ft documentation.", "human_check": "Run tests; inspect security, edge cases, correctness.",
"skills": "Python, debugging, testing, Git, prompt design." }, "2": { "role": "Teacher / Trai
ner", "task": "Drafting a lesson outline or quiz", "category": "Often assistive", "ai_help":
"Generate examples, question drafts, alternative explanations.", "human_check": "Check curric
ulum alignment, answer accuracy, accessibility.", "skills": "Subject knowledge, instructional
design, evaluation." }, "3": { "role": "Customer Support", "task": "Drafting a routine repl
y", "category": "Assistive with review", "ai_help": "Summarise a case and propose a response
draft.", "human_check": "Check policy, tone, personal data, and escalation needs.", "skills":
"Communication, product knowledge, empathy, privacy." } } def show_task(key, item): print
(f"\nRole: {item['role']}") print(f"Task: {item['task']}") print(f"Category: {item['categor
y']}") print(f"AI Assistance: {item['ai_help']}") print(f"Human Check: {item['human_chec
k']}") print(f"Skills to Build: {item['skills']}") def main(): print("=== GENAI JOB TASK EXPL
ORER ===") for key, item in TASKS.items(): print(f"{key}. {item['role']} - {item['task']}") c
hoice = input("\nEnter choice (1-3): ").strip() if choice in TASKS: show_task(choice, TASKS[c
hoice]) if __name__ == "__main__": main()
```

11. AI OUTPUT VERIFICATION & CORRECTION LOG

AI Claim to Test	Finding & Verification Correction
"AI will eliminate 1 in 4 jobs."	**Correction:** ILO (2025) states 1 in 4 workers have *exposure* to GenAI. Exposure does not equal job loss.
"AI creates 78 million jobs."	**Correction:** WEF projects 170M created & 92M displaced by 2030 across multiple market factors (net +78M).
"ChatGPT makes workers 40% faster."	**Correction:** Science (2023) study applied specifically to mid-level writing tasks, not all occupations.

12. RESEARCH WORKFLOW & PROMPT LIBRARY

- **Source Summary Prompt:** *"Summarise this report in 150 words. Preserve exact figures, dates, and caveats. Cite page numbers."*
- **Fact-Check Prompt:** *"Classify each sentence as supported or unsupported using only primary source documents."*
- **Code Review Prompt:** *"Review this Python code for syntax errors and edge-case handling without altering logic."*

13. REFERENCES & CONCLUSION

Conclusion: Generative AI acts primarily as a task-level augmentative tool rather than an immediate whole-job replacement engine. Students must focus on combining core technical capabilities with verification skills and critical thinking.

1. World Economic Forum (2025). *The Future of Jobs Report 2025*.
2. International Labour Organization (2025). *Generative AI and Jobs: A 2025 Update*. ILO Working Paper 140.
3. Noy, S., & Zhang, W. (2023). "Experimental evidence on the productivity effects of generative artificial intelligence." *Science*, 381(6654), 187–192.